28 *Use of the Data Booklet is relevant to this question.*

An electrochemical cell is set up using a $Zn^{2+}(aq)|Zn(s)$ half-cell and a $MnO_4^-(aq)|Mn^{2+}(aq)$ half-cell in acidic solution.

Which of the following gives a correct effect on the E_{cell} when each of the changes is made to the corresponding half-cell separately?

	change	half-cell	effect on E_{cell}
A	addition of water	$MnO_4^-(aq)\|Mn^{2+}(aq)$	less positive
B	addition of $Mn(NO_3)_2(s)$	$MnO_4^-(aq)\|Mn^{2+}(aq)$	no change
C	addition of $Zn(NO_3)_2(s)$	$Zn^{2+}(aq)\|Zn(s)$	more positive
D	addition of $Na_2CO_3(s)$	$Zn^{2+}(aq)\|Zn(s)$	no change